Notes: Fan (AER, 2013)

Ownership Consolidation and Product Characteristics: A Study of the US Daily Newspaper Market

Contents

1 Big picture **3**

2 Incremental contribution **3**

2.1 Contribution relative to standard merger simulation . . . 3

2.2 Contribution relative to endogenous-product-choice literature . . . 4

2.3 Contribution relative to newspaper and media-economics literature . . . 4

2.4 Contribution to identification . . . 4

2.5 Contribution to policy . . . 4

3 Data and measurement **4**

3.1 Newspaper demand data . . . 4

3.2 Subscription and advertising prices . . . 5

3.3 Newspaper content characteristics . . . 5

3.4 Demographics and market structure . . . 5

4 Models and empirical specifications **5**

4.1 Reader demand: multiple discrete choice . . . 5

4.2 Household indirect utility . . . 6

4.3 Content quality index . . . 6

4.4 Advertising demand . . . 6

4.5 Variable profit and costs . . . 6

4.6 First-stage characteristic choice . . . 7

4.7 Second-stage pricing conditions . . . 7

4.8 Estimation . . . 7

5 Identification and empirical design **7**

5.1 Endogeneity of prices and characteristics . . . 7

5.2 Partial-overlap instruments . . . 7

5.3 Why the instrument strategy is plausible . . . 8

5.4 What identifies main components . . . 8

5.5 Limitations . . . 8

6 Section-by-section study guide **8**

7 Figures and tables **9**

7.1 Figures 1–2: overlap and instruments . . . 9

7.2 Tables 1–3: data and instrument validity . . . 9

7.3 Table 4: estimation results . . . 10
7.4 Minneapolis case: Figure 3 and Tables 5–7 . . . 11
7.5 Cross-market simulations: Figures 4–5 and Table 8 . . . 12
7.6 Appendix Table A1: data sources . . . 13

8 Result map **14**

9 Short summary **14**

10 Conclusion **15**
10.1 Core conclusion . . . 15
10.2 Economic intuition . . . 15
10.3 Incremental contribution revisited . . . 15
10.4 Implications for antitrust . . . 15
10.5 Limitations . . . 15
10.6 Takeaway . . . 15

1 Big picture

- **Question.** How does ownership consolidation affect newspaper subscription prices, advertising rates, content quality, local news emphasis, variety, circulation, and welfare when publishers choose both prices and product characteristics?
- **Motivation.** Standard merger simulation often holds product characteristics fixed and studies only price changes. Fan argues this can miss a major channel of consumer harm because merging firms can also alter product quality, localness, and variety.
- **Setting.** The paper studies the US daily newspaper market from 1997 to 2005 using data on circulation, subscription prices, advertising rates, ad linage, ownership, newspaper staff, pages, publication frequency, and county demographics.
- **Two-sided market.** Newspapers sell copies to readers and advertising space to advertisers. Advertising demand depends on circulation, while reader demand depends on price and characteristics. Therefore price and quality choices have cross-side effects.
- **Key product characteristics.** The paper endogenizes three content characteristics: content quality, local news ratio, and content variety. Quality is built from news hole, opinion/editorial staff, and reporters. Local news ratio is the share of staff allocated to local news. Variety is one minus a concentration index over staff sections.
- **Identification idea.** Newspaper circulation areas partially overlap. Competitor demographics in nonoverlapping areas affect competitors' prices/characteristics but do not directly enter the focal newspaper's demand. This creates excluded instruments for endogenous prices and characteristics.
- **Main case study.** A counterfactual merger between the Minneapolis Star Tribune and the St. Paul Pioneer Press, which was blocked by the Department of Justice, predicts lower quality, lower local news ratio, lower variety, higher subscription prices, lower circulation, lower reader surplus, and higher publisher surplus.
- **Main quantitative result.** In the Minneapolis case, reader surplus falls by 3.28 million dollars and publisher surplus rises by 4.32 million dollars. Ignoring characteristic adjustment understates reader losses by 1.05 million dollars.
- **General result.** Across 2005 duopoly and triopoly markets, reader losses are larger in markets where households value newspapers more and where merging newspapers overlap more. Losses are smaller when the two merging newspapers are highly asymmetric in circulation.
- **Takeaway.** Merger analysis in media and other differentiated-product markets can be biased if it treats quality and product characteristics as fixed.

2 Incremental contribution

2.1 Contribution relative to standard merger simulation

- The paper endogenizes product characteristics in a structural merger model rather than holding quality fixed.
- It quantifies how much merger welfare analysis changes when the merged firm adjusts characteristics as well as prices.
- It shows that product adjustment is not a small detail in the newspaper market: in the Minneapolis case, characteristic changes account for roughly 40 percent of the total reader surplus loss.

2.2 Contribution relative to endogenous-product-choice literature

- Earlier empirical product-choice models often used discrete choice sets, monopoly settings, or profit functions not derived from demand.
- Fan estimates an oligopoly model in which publishers choose continuous product characteristics and the profit function is tied to estimated reader and advertising demand.
- The paper therefore bridges BLP-style demand estimation, two-sided markets, and continuous quality choice.

2.3 Contribution relative to newspaper and media-economics literature

- Prior newspaper studies often used reduced-form concentration or ownership measures. Fan explicitly models market structure, ownership, pricing, and product-characteristic choices.
- The paper creates a novel dataset that measures newspaper characteristics from staffing and page data, not just circulation or prices.
- It studies not only one merger but also the distribution of welfare effects across all duopoly and triopoly markets in the 2005 sample.

2.4 Contribution to identification

- Endogenous characteristics make classic BLP instruments problematic because competitors' characteristics are also choice variables.
- The paper's key instrument strategy uses demographics from competitors' markets that do not overlap with the focal newspaper's market.
- Partial overlap in circulation areas becomes both the core market-definition problem and the econometric solution.

2.5 Contribution to policy

- The paper shows how antitrust authorities can evaluate media mergers when firms adjust quality, local content, and variety.
- It shows that ignoring quality adjustment can underestimate harm to readers and over/underestimate publisher gains.
- It highlights that media mergers can affect public-good-like aspects of newspapers such as local news and content diversity.

3 Data and measurement

3.1 Newspaper demand data

- Circulation is measured at the newspaper-county-year level. For ABC newspapers, circulation comes from Newspaper County Circulation Reports. For non-ABC members, circulation comes from sworn postal statements.
- Markets are not single counties. A newspaper can circulate in many counties and can partially overlap with multiple other newspapers.
- The model uses county-level circulation because households in a county face a set of available newspapers.
- Some county-years have total newspaper circulation above the number of households, motivating the multiple-discrete-choice model and allowing duplicate readership.

Interpretation: The data require a model that can handle overlapping circulation areas and multi-homing by readers. A single-choice model would mechanically rule out households buying more than one newspaper.

3.2 Subscription and advertising prices

- Annual subscription prices and display advertising rates come from Editor and Publisher International Year Book.
- Display advertising rate is measured as dollars per column inch.
- Advertising quantity is annual display ad linage, from TNS Media Intelligence for a subsample of newspaper-years.
- The paper separately models display advertising demand but does not model preprint advertising demand because the required preprint rates are not observed.

Interpretation: Newspaper revenue is two-sided. A publisher may accept lower circulation revenue or alter quality because circulation raises advertising demand.

3.3 Newspaper content characteristics

- **Content quality:** a weighted index of nonadvertising space, opinion/editorial staff, and reporters.
- **Local news ratio:** the share of staff assigned to local news.
- **Variety:** $100[1 - \sum_s share_s^2]$, where $share_s$ is the staff share in section s.
- Staff counts are built from Bacon's Newspaper Directory. A staff member with multiple titles is weighted to avoid double-counting.

Interpretation: These variables are not perfect measures of journalistic quality, but they are much richer than simply observing number of pages or circulation.

3.4 Demographics and market structure

- County demographics include education, income, age, urbanization, and number of households.
- Distance from county center to newspaper home county captures local preference for nearby newspapers.
- Ownership is measured by publisher. Ownership determines which newspapers' profits are jointly internalized.
- Partial overlap indexes summarize how much newspaper circulation areas overlap.

Interpretation: The market structure is continuous and geographically irregular. This motivates a structural model rather than a simple concentration index.

4 Models and empirical specifications

4.1 Reader demand: multiple discrete choice

A household can choose zero, one, or two newspapers. The probability a household buys newspaper j is the probability j is the first-best choice plus the probability that another newspaper is first-best but j is the second-best newspaper and gives enough additional utility after a diminishing-utility penalty κ.

$$Pr(j \text{ chosen}) = Pr(j \text{ first choice}) + Pr(j \text{ second choice and } u_{ijct} - \kappa \geq u_{i0ct}).$$

Interpretation: The parameter κ captures the loss from buying a second newspaper. If κ were infinite, the model would collapse to a standard single-discrete-choice model.

4.2 Household indirect utility

$$u_{ijct} = \alpha p_{jt} + x_{jt}\beta_{ict} + y_{jct}\psi + z_{ct}\phi + \xi_{jct} + \varepsilon_{ijt}.$$

- p_{jt} is subscription price.
- x_{jt} contains endogenous characteristics: quality, local news, and variety.
- y_{jct} contains exogenous product-county variables such as distance.
- z_{ct} contains county demographics.
- ξ_{jct} is unobserved product-county-year quality.

Interpretation: Price and characteristics are endogenous because publishers choose them while observing demand and cost shocks.

4.3 Content quality index

$$x_{1jt} = x_{11jt} + \omega_2 x_{12jt} + \omega_3 x_{13jt}.$$

Here x_{11jt} is news hole, x_{12jt} is opinion/editorial staff, and x_{13jt} is reporters. Local news ratio is x_{2jt} and variety is $x_{3jt} = 100[1 - \sum_s share_{sjt}^2]$.

Interpretation: The quality index combines space and staff. Variety is a diversity measure analogous to one minus a Herfindahl index.

4.4 Advertising demand

$$a(r_{jt}, q_{jt}, H_{jt}; \eta, \lambda) = \exp(\eta) H_{jt}^{\lambda_0} q_{jt}^{\lambda_1} r_{jt}^{\lambda_2}.$$

Equivalently,

$$\log a_{jt} = \eta + \lambda_0 \log H_{jt} + \lambda_1 \log q_{jt} + \lambda_2 \log r_{jt} + \iota_{jt}.$$

Interpretation: Advertising demand rises with market size and circulation and falls with advertising rates. The circulation elasticity is especially important because it links reader-side choices to advertiser-side profit.

4.5 Variable profit and costs

$$\pi_j^{II} = (p_j q_j - ac_j^{(q)} q_j) + (r_j a_j - mc_j^{(a)} a_j) + \mu_1 q_j + \frac{1}{2}\mu_2 q_j^2.$$

Circulation cost allows economies of scale and scope through total circulation Q_j of commonly owned nearby newspapers:

$$ac_j^{(q)} = (\gamma_1 + \gamma_2 f_j + \gamma_3 (x_{1j} + a_j)) \log(Q_j)^{\gamma_4} + \omega_j.$$

Interpretation: The cost structure permits mergers to lower printing and delivery costs. This cost synergy interacts with subscription prices and content choices.

4.6 First-stage characteristic choice

The first-stage profit is

$$\pi_j^I(x) = \pi_j^{II}(p^*(x), r^*(x); x) - fc(x_j, \nu_j; \tau).$$

Publishers choose characteristics anticipating the equilibrium price and advertising-rate responses.

Interpretation: Ownership consolidation changes the objective function because the merged publisher internalizes cross-effects of each newspaper's characteristics and prices on the other newspaper's profits.

4.7 Second-stage pricing conditions

The advertising rate first-order condition is

$$r_{jt} = \bar{\zeta} + \frac{\gamma_3}{1 + 1/\lambda_2} \log(Q_{jt})^{\gamma_4} q_{jt} + \zeta_{jt}.$$

The subscription-price condition can be written in matrix form:

$$p = \Delta^{-1} q - [\Lambda + (\mu_1 + \mu_2 q)] + \Gamma q + ac^{(q)}.$$

Interpretation: The price equation differs from a one-sided pricing equation because circulation affects advertising profit, preprint profit, and economies of scale/scope.

4.8 Estimation

Five equations are taken to the data: newspaper demand, advertising demand, advertising-rate FOC, subscription-price FOC, and characteristic FOC.

$$\{(5), (8), (16), (17), (18)\}.$$

Estimation uses GMM and exploits excluded instruments from competitors' nonoverlapping demographics.

Interpretation: The model is estimated jointly so that the counterfactual simulations can adjust prices, ad rates, and characteristics consistently.

5 Identification and empirical design

5.1 Endogeneity of prices and characteristics

Prices and product characteristics are chosen by publishers and are likely correlated with unobserved demand and cost shocks. Standard product-characteristic instruments are invalid because competitors' characteristics are endogenous.

5.2 Partial-overlap instruments

The excluded instruments are demographics in competitors' circulation areas that do not overlap with the focal newspaper's own circulation area. These demographics affect competitors' choices, which affect the focal newspaper through strategic interaction, but do not directly enter the focal newspaper's demand in its own counties.

5.3 Why the instrument strategy is plausible

- The newspaper industry naturally has partially overlapping circulation areas.
- Demand shifters in a competitor-only area affect the competitor's characteristics and pricing.
- Through oligopolistic competition, those choices affect the focal newspaper's optimal choices.
- Because the area is nonoverlapping, the demographics do not directly determine the focal newspaper's county demand.

Interpretation: Figure 1 and Figure 2 are essential because they make the exclusion logic visual: the geographic structure creates variation not usually available in product-choice models.

5.4 What identifies main components

- Reader demand is identified by circulation variation across counties, prices, characteristics, demographics, and distance.
- Advertising demand is identified by advertising linage, ad rates, circulation, and market size.
- Cost parameters are identified by first-order conditions for subscription price, advertising rate, and characteristics.
- The diminished utility of the second newspaper is identified by variation in the number of available newspapers and duplicate readership.

5.5 Limitations

- Preprint demand is not modeled because the required preprint advertising rates are unavailable.
- Newspaper exit is not modeled because exits are infrequent over the sample period.
- Content characteristics are measured through staffing and pages, which are proxies rather than direct measures of article quality.
- The model assumes equilibrium and uses best-response iteration for simulations.

6 Section-by-section study guide

- **Introduction.** Establishes why product-characteristic adjustment matters for merger welfare analysis and previews the Minneapolis counterfactual.
- **Model: demand.** Develops a multiple-discrete-choice model that allows a household to buy two newspapers and includes price, quality, local news, variety, demographics, and distance.
- **Model: supply.** Derives publisher profit from circulation, display advertising, preprint profit, cost synergies, and characteristic fixed costs.
- **Estimation.** Explains GMM estimation and the nonoverlap demographic instruments.
- **Minneapolis merger.** Simulates the blocked Star Tribune/Pioneer Press merger with and without characteristic adjustment.
- **Cross-market simulations.** Studies welfare effects in all duopoly and triopoly markets and links welfare effects to penetration, overlap, asymmetry, and elasticity.
- **Conclusion.** Summarizes policy lessons and limitations.

7 Figures and tables

7.1 Figures 1–2: overlap and instruments

Read: Figure 1 shows that newspapers' circulation areas partially overlap; Figure 2 shows how nonoverlapping competitor demographics can instrument for endogenous prices and characteristics.

Detailed interpretation: Partial overlap is the paper's core empirical opportunity. If circulation areas did not overlap, there would be no local competition to exploit. If they overlapped perfectly, competitor demographics would directly affect the focal newspaper's demand. Partial overlap creates variation that affects competitors but is excluded from the focal paper's demand.

Economic message: The same geography that complicates market definition solves the instrument problem.

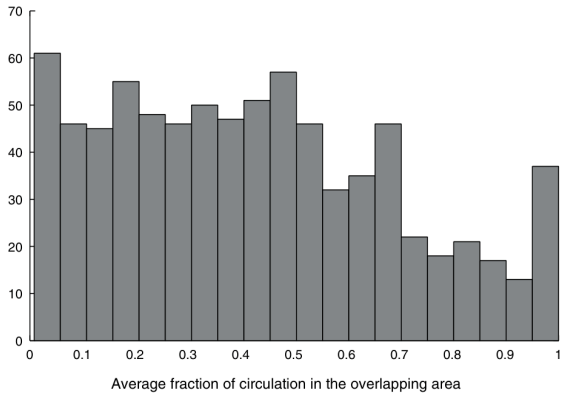

FIGURE 1. NEWSPAPER COVERAGE OVERLAP IN 2005

(a) Figure 1: Newspaper Coverage Overlap in 2005

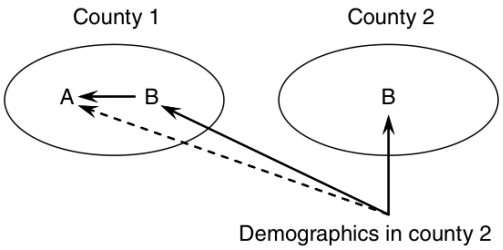

FIGURE 2. INSTRUMENTS

(b) Figure 2: Instruments

7.2 Tables 1–3: data and instrument validity

Read: Table 1 summarizes player newspapers; Table 2 summarizes county demographics; Table 3 reports correlations between included and excluded demographic instruments.

Detailed interpretation: Table 1 shows the enormous scale and heterogeneity of newspapers: mean circulation is 19,117, mean subscription price is 101.24 dollars, mean local news ratio is 13.69 percent, and mean variety is 69.80. Table 2 documents demographic variation across counties. Table 3 shows that nonoverlapping demographics are not mechanically identical to own-market demographics, supporting instrument relevance and exclusion logic.

Economic message: The empirical design uses rich cross-sectional variation in local markets, not just time-series variation in ownership.

Table 1—Summary Statistics of Player Newspapers

	Mean	Median	SD	Min	Max	Observations
Total circulation	19,117	9,322	32,739	1,132	468,483	5,843[a]
Annual display ad linage (column inch)	1,362,154	1,168,452	760,828	123,124	4,337,941	422
Price of newspapers ($)	101.24	96.61	33.75	13.19	365.31	5,843
Display advertising rate ($/column inch)	23.50	12.91	36.16	2.45	520.29	5,843
Frequency (issues/52 weeks)	308.16	312.00	53.70	208.00	364.00	5,843
Pages (pages/issue)	27.06	22.86	16.47	8.00	167.43	5,843
Opinion staff	1.36	0.33	3.05	0	42.42	5,843
Reporters	5.31	2.00	12.44	0	210.00	5,843
Local news ratio (percent)	13.69	13.89	10.16	0	62.50	5,843
Variety	69.80	75.06	17.50	0	88.63	5,843
Market penetration (percent)	33.70	32.82	16.63	0.56	97.08	11,203[b]
Distance to home county (100km)	0.26	0	0.38	0	3.95	11,203

(a) Table 1: Summary Statistics of Player Newspapers

Table 2—Summary Statistics: County Demographics

	Mean	Median	SD	Min	Max	Observations
Percent of population over 25 with bachelor's degree or higher	16.88	15.05	6.95	5.64	60.48	8,947[c]
Median income ($1,000)	34.17	32.82	7.09	16.36	76.31	
Median age	36.56	36.70	3.73	20.70	54.30	
Urbanization (percent)	48.80	50.50	26.05	0	100	
Number of households	28,550	14,876	47,949	710	858,719	

(b) Table 2: Summary Statistics: County Demographics

Table 3—Correlation of Demographics in Neighboring Counties

	Educational level	Median income	Median age	Urbanization
Correlation	0.1277	0.3509	0.4526	0.2767

(c) Table 3: Correlation of Demographics in Neighboring Counties

7.3 Table 4: estimation results

Read: Table 4 reports reader demand, advertising demand, cost, and preprint-profit estimates.

Detailed interpretation: Reader demand is price-sensitive and positively values content quality and variety. The estimated diminishing utility parameter implies duplicate readership is limited in many county-years but still important enough to model. Advertising demand increases with market size and circulation and decreases with advertising rates. Cost estimates imply economies of scale/scope in printing and delivery; the text reports a cost saving of about 27 percent when equal-size nearby newspapers with equal pages/frequency merge and circulation is simply summed, or 17 percent if postmerger circulation is 75 percent of the total premerger circulation.

Economic message: Table 4 provides the structural primitives that drive the counterfactual simulations: consumer valuations, advertiser responses, and publisher cost incentives.

Table 4—Estimation Results

	Parameter	Estimate	SE
Utility			
Price ($100)	α	−1.411***	0.030
log(1 + content quality)	β_1	0.772***	0.017
log(1 + local news)	β_2	−0.002***	0.001
log(1 + variety)	β_3	0.013***	0.001
Weight on news hole in content quality	1		
Weight on opinion in content quality	ϖ_2	0.466*	0.301
Weight on reporters in content quality	ϖ_3	6.264***	0.322
log(1 + local news), education	θ_1	0.003***	0.000
log(1 + local news), median age	θ_2	0.004***	0.001
log(1 + local news), standard deviation	σ	0.007	0.016
log(households in the market)	ψ_1	−0.728***	0.015
Morning edition	ψ_2	0.072***	0.008
Local dummy	ψ_3	0.541***	0.007
County distance (1,000km)	ψ_4	−3.904***	0.061
Constant	φ_0	4.990***	0.130
Education	φ_1	0.607***	0.039
Median income ($10,000)	φ_2	−0.203***	0.037
Median age	φ_3	0.043***	0.001
Urbanization	φ_4	0.640***	0.016
Time	ρ	0.144***	0.002
Diminishing utility	κ	1.919***	0.078
Display advertising demand			
Ad market size	λ_0	0.043**	0.026
Total circulation	λ_1	1.673***	0.039
Ad rate	λ_2	−1.195***	0.029
Constant	ϕ	−1.625***	0.269
Average cost of circulation			
Constant	γ_1	−82.315	145.240
Frequency	γ_2	1.768***	0.112
Pages in a year	γ_3	0.022***	0.004
Economies of scale/scope	γ_4	−0.448***	0.045
Marginal cost of ad sales	$\bar{\zeta}$	11.775***	0.518
Slope of the fixed cost for content quality			
Constant	τ_{10}	91,248***	2,932
Content quality	τ_{11}	0.00003	17.929[a]
Slope of the fixed cost for local news			
Constant	τ_{20}	88.681	181.590
Local news	τ_{21}	4.650**	2.803
Slope of the fixed cost for variety			
Constant	τ_{30}	1.276	68.278
Variety	τ_{31}	0.373	2.511
Preprint profit			
Circulation	μ_1	142.690***	54.610

Table 4: Estimation Results

7.4 Minneapolis case: Figure 3 and Tables 5–7

Read: Figure 3 maps the designated markets around the Star Tribune, Pioneer Press, and other local newspapers. Table 5 simulates the merger holding characteristics fixed. Table 6 allows prices and characteristics to adjust. Table 7 reports welfare effects.

Detailed interpretation: Without characteristic adjustment, the simulation captures only price and ad-rate effects. With characteristic adjustment, both the Star and Pioneer reduce content quality, local news ratio, and variety, while subscription prices rise. Overall local news content read per household falls 10.75 percent. Table 7 shows reader surplus falls by 3.28 million dollars, advertiser surplus falls by 7.10 percent, and publisher surplus rises by 4.32 million dollars. Ignoring characteristic adjustment understates reader loss by 1.05 million dollars and publisher gains by 0.10 million dollars.

Economic message: The merger changes the newspaper product itself. Antitrust analysis that holds characteristics fixed misses an important part of consumer harm.

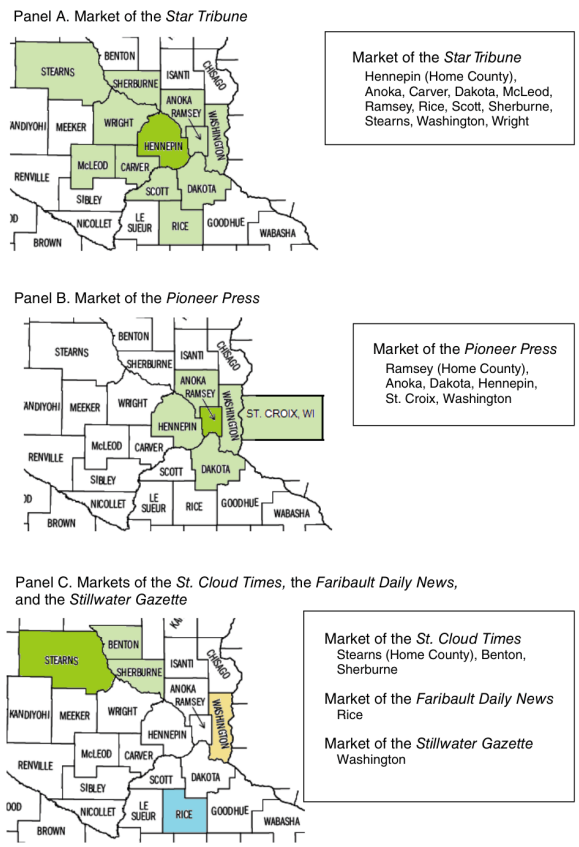

FIGURE 3. NEWSPAPER DESIGNATED MARKET

(a) Figure 3: Newspaper Designated Market

TABLE 5—EFFECTS OF OWNERSHIP CONSOLIDATION OF THE *STAR* AND THE *PIONEER* WITHOUT CHARACTERISTIC ADJUSTMENT

		Star	Pioneer	Faribault	St. Cloud	Stillwater
Price ($/year)	before	172.79	171.51	111.31	150.07	78.33
	after	175.98	179.52	111.32	149.95	75.03
	change	3.19	8.01	0	−0.12	−3.30
Ad rate ($/column inch)	before	230.88	153.08	12.37	44.15	11.13
	after	227.00	147.07	12.39	44.19	11.25
	change	−3.87	−6.00	0.02	0.03	0.12
Circulation	before	317,337	159,864	6,384	24,578	3,341
	after	310,148	148,519	6,434	24,667	3,644
	change	−7,189	−11,345	50	89	303

(b) Table 5: Effects Without Characteristic Adjustment

AND THE *PIONEER* WITH CHARACTERISTIC ADJUSTMENT

		Star	Pioneer	Faribault	St. Cloud	Stillwater
Content quality index	before	788.49	474.29	7.00	65.28	0.70
	after	771.78	422.59	7.17	66.26	0.31
	change	−16.72	−51.70	0.17	0.98	−0.40
Local news (percent)	before	22.00	27.48	14.29	35.42	0
	after	21.15	23.88	14.47	35.60	0.05
	change	−0.85	−3.60	0.18	0.18	0.05
Variety	before	83.38	82.07	50.00	74.50	0
	after	81.79	74.61	50.35	75.01	0
	change	−1.58	−7.46	0.35	0.51	0
Price ($/year)	before	172.79	171.51	111.31	150.07	78.33
	after	175.39	178.83	111.26	149.64	87.41
	change	2.59	7.32	−0.05	−0.43	9.08
Ad rate ($/column inch)	before	230.88	153.08	12.37	44.15	11.13
	after	227.09	144.4	12.42	44.29	10.83
	change	−3.79	−8.68	0.05	0.13	−0.3
Circulation	before	317,337	159,864	6,384	24,578	3,341
	after	310,223	140,635	6,518	24,939	2,597
	change	−7,114	−19,229	134	361	−744

(a) Table 6: Effects With Characteristic Adjustment

TABLE 7—WELFARE EFFECTS OF OWNERSHIP CONSOLIDATION OF THE *STAR* AND THE *PIONEER*

	Without characteristic adjustment	With characteristic adjustment
Change in *RS* (million $)	−2.22	−3.28
Percentage change in *RS*	−4.67	−6.87
Percentage change in *AS*	−4.66	−7.10
Change in *PS* (million $)	4.23	4.32
Percentage change in *PS*	36.41	37.25

(b) Table 7: Welfare Effects of Star/Pioneer Consolidation

7.5 Cross-market simulations: Figures 4–5 and Table 8

Read: Figures 4 and 5 show the distribution of welfare effects for duopoly and triopoly mergers, with and without characteristic adjustment. Table 8 reports the mean effects.

Detailed interpretation: In Table 8, duopoly mergers with characteristic adjustment reduce average per-household reader surplus by 6.83 dollars and reduce total reader surplus by 0.36 million dollars on average, while raising publisher surplus by 0.22 million dollars. Triopoly mergers with characteristic adjustment

reduce total reader surplus by 0.93 million dollars and raise publisher surplus by 0.69 million dollars on average. Ignoring characteristic adjustment substantially understates losses and gains, especially in triopoly markets.

Economic message: Characteristic adjustment matters in many markets, not just the Minneapolis case. The bias from ignoring quality choice is systematic and market-structure dependent.

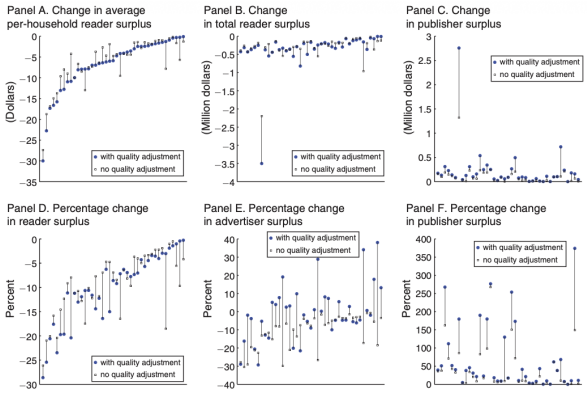

Figure 4. Welfare Implications of Duopoly Mergers

(a) Figure 4: Welfare Implications of Duopoly Mergers

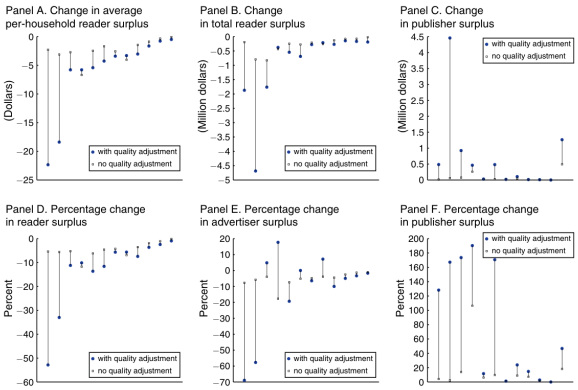

Figure 5. Welfare Implications of Triopoly Mergers

(b) Figure 5: Welfare Implications of Triopoly Mergers

Table 8—Mean Welfare Changes across Duopoly and Triopoly Markets

	$\overline{\Delta RS}$ (\$)	ΔRS (millions \$)	ΔPS (millions \$)	$\%\Delta RS$ (percent)	$\%\Delta AS$ (percent)	$\%\Delta PS$ (percent)
Duopoly with characteristic adjustment	−6.83	−0.36	0.22	−10.17	−0.62	70.05
Duopoly without characteristic adjustment	−6.28	−0.31	0.11	−9.33	−10.92	38.91
Triopoly with characteristic adjustment	−6.22	−0.93	0.69	−13.19	−11.9	77.55
Triopoly without characteristic adjustment	−2.35	−0.29	0.08	−4.71	−5.48	14.96

Table 8: Mean Welfare Changes across Duopoly and Triopoly Markets

7.6 Appendix Table A1: data sources

Read: Table A1 lists the variables, data descriptions, and sources used for newspaper demand, advertising demand, prices, characteristics, ownership, distance, and demographics.

Detailed interpretation: The data infrastructure is a central contribution. The paper combines ABC circulation, SRDS circulation, TNS advertising linage, E&P prices and pages, Bacon staffing data, publisher ownership, Census demographics, and county geography. The model is only possible because these sources are linked at newspaper-county-year and newspaper-year levels.

Economic message: The dataset makes a price-plus-quality merger analysis feasible.

Table A1—Data Description and Sources

	Var	Data description	Data source
Newspaper demand	q_{jct}	County circulation	ABC, SRDS
Display ad demand	a_{jt}	Annual display advertising linage	TNS
Price of newspaper	p_{jt}	Annual subscription price (1997 \$)	E&P
Price of display ad	r_{jt}	Adverting rate (1997 \$/column inch)	E&P
Newspaper characteristics	x_{12jt}	Weighted count of columnists and editorial editors	Bacon
	x_{13jt}	Weighted count of reporters and correspondents	Bacon
	x_{2jt}	Share of staff on local news	Bacon
	x_{3jt}	$100\left[1 - \sum_i (\text{share of staff in section } i)^2\right]$	Bacon
	f_{jt}	Frequency of publication (issues/52 weeks)	E&P
	y_{2jt}	Edition (morning or evening)	E&P
	n_{jt}	Annual number of pages	E&P
County distance	y_{4jct}	Distance between county c and newspaper j's home county	E&P, census
Ownership		Publisher	Bacon
County demographics	z_{1c}	Percent of population over 25 with bachelor's degree or higher	Census
	z_{2c}	Median income (1997 \$)	Census
	z_{3c}	Median age	Census
	z_{4c}	Percent of urban population	Census
	H_{ct}	Number of households	ABC

Notes: ABC: County Circulation Report by Audit Bureau of Circulations; Bacon: Bacon's Newspaper Directory; E&P: Editor and Publisher International Year Book; SRDS: SRDS Circulation; TNS: TNS Media Intelligence.

Table A1: Data Description and Sources

8 Result map

- **Identification result.** Partial overlap of circulation areas creates excluded instruments based on competitors' nonoverlapping demographics.
- **Demand result.** Readers dislike high prices and value newspaper characteristics, especially content quality and variety.
- **Advertising result.** Advertising demand is increasing in market size and circulation and decreasing in advertising rate; circulation elasticity is larger than one.
- **Cost result.** Printing and delivery have scale/scope economies, so merged ownership can lower costs.
- **Minneapolis result.** The blocked Star/Pioneer merger would raise subscription prices and reduce quality, localness, and variety.
- **Welfare result.** Reader surplus losses are larger when characteristic adjustment is allowed; publisher surplus gains are also larger.
- **Cross-market result.** Welfare losses are larger with high newspaper penetration and high overlap, and smaller when merging papers are highly asymmetric.
- **Policy result.** Price-only merger simulations can understate consumer harm in media markets.

9 Short summary

- Fan estimates a structural model of the US daily newspaper market with endogenous prices and product characteristics.
- The model includes reader demand, advertising demand, pricing, advertising rates, and publisher choices of content quality, local news ratio, and variety.
- The key identification strategy uses competitors' nonoverlapping market demographics as instruments.

- The paper uses a new dataset for 1997-2005 newspaper circulation, prices, advertising, staffing, pages, ownership, and demographics.
- In the simulated Minneapolis merger, both newspapers reduce characteristics and raise prices.
- Reader surplus falls by 3.28 million dollars and publisher surplus rises by 4.32 million dollars.
- Ignoring characteristic adjustment understates reader losses by 1.05 million dollars.
- Across duopoly and triopoly mergers, welfare effects depend on penetration, overlap, asymmetry, and elasticity.
- The key lesson is that merger analysis should allow firms to reposition or degrade product characteristics.

10 Conclusion

10.1 Core conclusion

Fan's central conclusion is that ownership consolidation changes both price and product characteristics. In the newspaper market, mergers can reduce content quality, local news, and variety while raising subscription prices. Ignoring the product-characteristic channel can materially understate consumer harm.

10.2 Economic intuition

After a merger, the publisher internalizes substitution between commonly owned newspapers. This reduces the incentive to maintain high characteristics on both products and changes the optimal price-quality bundle. At the same time, circulation affects advertising demand and advertising revenue, so the merged firm balances reader-side and advertiser-side incentives.

10.3 Incremental contribution revisited

The paper's incremental contribution is to make quality adjustment central to structural merger analysis. It combines a two-sided newspaper model, continuous product-characteristic choices, new staffing-based quality measures, and a partial-overlap instrument strategy. This combination allows the paper to quantify welfare bias from holding product characteristics fixed.

10.4 Implications for antitrust

Antitrust analysis should consider not only price increases but also quality degradation, loss of local content, and reduced variety. These dimensions are especially important in media markets because local information and content diversity have public significance beyond private subscription value.

10.5 Limitations

The model does not estimate preprint advertising demand, cannot model newspaper shutdowns, and uses staffing/page measures as proxies for content. The counterfactual assumes the estimated equilibrium model is valid after ownership changes. These limitations matter, but they do not overturn the main policy lesson that quality adjustment can be large.

10.6 Takeaway

Price-only merger simulation is incomplete when firms can choose product characteristics. In the US daily newspaper market, ownership consolidation can reduce content quality and local news in ways that materially

change welfare analysis.